Mode 12 — Baseline-BFP

C-DP4 (C16×C16) → DP8 (16×16) → DP8 (16×8) → DP8 (16×4) → DP8 (8×4) → block exponents

Step 1 — C-DP4 (C16×C16)

$$\text{C-DP4}\,(\text{C16}\times\text{C16}) = D = \sum_{i=0}^{3} a_i\, b_i, \qquad a_i = a_i^{re} + \mathrm{j}\, a_i^{im}, \quad b_i = b_i^{re} + \mathrm{j}\, b_i^{im}$$

Step 2 — DP8 (16×16)

$$\mathrm{Re}(D) = \underbrace{\sum_{i=0}^{3} a_i^{re} b_i^{re} - \sum_{i=0}^{3} a_i^{im} b_i^{im}}_{\text{DP8 (16×16)}}$$

$$\mathrm{Im}(D) = \underbrace{\sum_{i=0}^{3} a_i^{re} b_i^{im} + \sum_{i=0}^{3} a_i^{im} b_i^{re}}_{\text{DP8 (16×16)}}$$

Step 3 — DP8 (16×8) **(byte-split $b = 2^8 b^{hi} + b^{lo}$)**

$$\mathrm{Re}(D) = \underbrace{2^8 \sum_{i=0}^{3} a_i^{re} b_i^{re,hi} - 2^8 \sum_{i=0}^{3} a_i^{im} b_i^{im,hi}}_{\text{DP8 (16×8)}} + \underbrace{\sum_{i=0}^{3} a_i^{re} b_i^{re,lo} - \sum_{i=0}^{3} a_i^{im} b_i^{im,lo}}_{\text{DP8 (16×8)}}$$

$$\mathrm{Im}(D) = \underbrace{2^8 \sum_{i=0}^{3} a_i^{re} b_i^{im,hi} + 2^8 \sum_{i=0}^{3} a_i^{im} b_i^{re,hi}}_{\text{DP8 (16×8)}} + \underbrace{\sum_{i=0}^{3} a_i^{re} b_i^{im,lo} + \sum_{i=0}^{3} a_i^{im} b_i^{re,lo}}_{\text{DP8 (16×8)}}$$

Step 4 — DP8 (16×4) **(nibble-split each byte)**

$$\begin{aligned}
\mathrm{Re}(D) = {} & \underbrace{2^{12} \sum_{i=0}^{3} a_i^{re} b_i^{re,hh} - 2^{12} \sum_{i=0}^{3} a_i^{im} b_i^{im,hh}}_{\text{DP8 (16×4)}} \\
& + \underbrace{2^{8} \sum_{i=0}^{3} a_i^{re} b_i^{re,hl} - 2^{8} \sum_{i=0}^{3} a_i^{im} b_i^{im,hl}}_{\text{DP8 (16×4)}} \\
& + \underbrace{2^{4} \sum_{i=0}^{3} a_i^{re} b_i^{re,lh} - 2^{4} \sum_{i=0}^{3} a_i^{im} b_i^{im,lh}}_{\text{DP8 (16×4)}} \\
& + \underbrace{\sum_{i=0}^{3} a_i^{re} b_i^{re,ll} - \sum_{i=0}^{3} a_i^{im} b_i^{im,ll}}_{\text{DP8 (16×4)}}
\end{aligned}$$

$$\begin{aligned}
\mathrm{Im}(D) = {} & \underbrace{2^{12} \sum_{i=0}^{3} a_i^{re} b_i^{im,hh} + 2^{12} \sum_{i=0}^{3} a_i^{im} b_i^{re,hh}}_{\text{DP8 (16×4)}} \\
& + \underbrace{2^{8} \sum_{i=0}^{3} a_i^{re} b_i^{im,hl} + 2^{8} \sum_{i=0}^{3} a_i^{im} b_i^{re,hl}}_{\text{DP8 (16×4)}}
\end{aligned}$$

$$+ \underbrace{2^4 \sum_{i=0}^{3} a_i^{re} b_i^{im,lh} + 2^4 \sum_{i=0}^{3} a_i^{im} b_i^{re,lh}}_{\text{DP8 (16×4)}}$$

$$+ \underbrace{\sum_{i=0}^{3} a_i^{re} b_i^{im,ll} + \sum_{i=0}^{3} a_i^{im} b_i^{re,ll}}_{\text{DP8 (16×4)}}$$

Step 5 — DP8 (8×4) **(byte-split $a = 2^8 a^{hi} + a^{lo}$)**

$$\mathrm{Re}(D) = \underbrace{2^{20} \sum_{i=0}^{3} a_i^{re,hi} b_i^{re,hh} - 2^{20} \sum_{i=0}^{3} a_i^{im,hi} b_i^{im,hh}}_{\text{DP8 (8×4)}}$$

$$+ \underbrace{2^{12} \sum_{i=0}^{3} a_i^{re,lo} b_i^{re,hh} - 2^{12} \sum_{i=0}^{3} a_i^{im,lo} b_i^{im,hh}}_{\text{DP8 (8×4)}}$$

$$+ \underbrace{2^{16} \sum_{i=0}^{3} a_i^{re,hi} b_i^{re,hl} - 2^{16} \sum_{i=0}^{3} a_i^{im,hi} b_i^{im,hl}}_{\text{DP8 (8×4)}}$$

$$+ \underbrace{2^{8} \sum_{i=0}^{3} a_i^{re,lo} b_i^{re,hl} - 2^{8} \sum_{i=0}^{3} a_i^{im,lo} b_i^{im,hl}}_{\text{DP8 (8×4)}}$$

$$+ \underbrace{2^{12} \sum_{i=0}^{3} a_i^{re,hi} b_i^{re,lh} - 2^{12} \sum_{i=0}^{3} a_i^{im,hi} b_i^{im,lh}}_{\text{DP8 (8×4)}}$$

$$+ \underbrace{2^{4} \sum_{i=0}^{3} a_i^{re,lo} b_i^{re,lh} - 2^{4} \sum_{i=0}^{3} a_i^{im,lo} b_i^{im,lh}}_{\text{DP8 (8×4)}}$$

$$+ \underbrace{2^{8} \sum_{i=0}^{3} a_i^{re,hi} b_i^{re,ll} - 2^{8} \sum_{i=0}^{3} a_i^{im,hi} b_i^{im,ll}}_{\text{DP8 (8×4)}}$$

$$+ \underbrace{\sum_{i=0}^{3} a_i^{re,lo} b_i^{re,ll} - \sum_{i=0}^{3} a_i^{im,lo} b_i^{im,ll}}_{\text{DP8 (8×4)}}$$

$$\mathrm{Im}(D) = \underbrace{2^{20} \sum_{i=0}^{3} a_i^{re,hi} b_i^{im,hh} + 2^{20} \sum_{i=0}^{3} a_i^{im,hi} b_i^{re,hh}}_{\text{DP8 (8×4)}}$$

$$+ \underbrace{2^{12} \sum_{i=0}^{3} a_i^{re,lo} b_i^{im,hh} + 2^{12} \sum_{i=0}^{3} a_i^{im,lo} b_i^{re,hh}}_{\text{DP8 (8×4)}}$$

$$+ \underbrace{2^{16} \sum_{i=0}^{3} a_i^{re,hi} b_i^{im,hl} + 2^{16} \sum_{i=0}^{3} a_i^{im,hi} b_i^{re,hl}}_{\text{DP8 (8×4)}}$$

$$+\underbrace{2^{8}\sum_{i=0}^{3}a_i^{re,lo}b_i^{im,hl}+2^{8}\sum_{i=0}^{3}a_i^{im,lo}b_i^{re,hl}}_{\text{DP8 (8×4)}}$$

$$+\underbrace{2^{12}\sum_{i=0}^{3}a_i^{re,hi}b_i^{im,lh}+2^{12}\sum_{i=0}^{3}a_i^{im,hi}b_i^{re,lh}}_{\text{DP8 (8×4)}}$$

$$+\underbrace{2^{4}\sum_{i=0}^{3}a_i^{re,lo}b_i^{im,lh}+2^{4}\sum_{i=0}^{3}a_i^{im,lo}b_i^{re,lh}}_{\text{DP8 (8×4)}}$$

$$+\underbrace{2^{8}\sum_{i=0}^{3}a_i^{re,hi}b_i^{im,ll}+2^{8}\sum_{i=0}^{3}a_i^{im,hi}b_i^{re,ll}}_{\text{DP8 (8×4)}}$$

$$+\underbrace{\sum_{i=0}^{3}a_i^{re,lo}b_i^{im,ll}+\sum_{i=0}^{3}a_i^{im,lo}b_i^{re,ll}}_{\text{DP8 (8×4)}}$$

Step 6 — Block exponents (one exponent block per DP8 (8×4) — all copies of one E; re/im and both B arrangements tied, negation scale-blind)

$$\big(P_j,\ E\big)\ \text{ per DP8 (8×4) above,}\qquad E=E_a+E_b\ \longrightarrow\ \big(M^{re},\ E\big),\quad \big(M^{im},\ E\big)$$

$\delta\equiv 0$: M^{re}, M^{im} are the Step-5 sums unchanged; $\operatorname{Re}(\hat{D})=M^{re}\cdot 2^{E}$, $\operatorname{Im}(\hat{D})=M^{im}\cdot 2^{E}$.

Step 7 — Result

$$\boxed{\begin{aligned}
M^{re}=\;&2^{20}\Big(\sum_{i=0}^{3}a_i^{re,hi}b_i^{re,hh}-\sum_{i=0}^{3}a_i^{im,hi}b_i^{im,hh}\Big)+2^{16}\Big(\sum_{i=0}^{3}a_i^{re,hi}b_i^{re,hl}-\sum_{i=0}^{3}a_i^{im,hi}b_i^{im,hl}\Big)\\
&+2^{12}\Big(\sum_{i=0}^{3}a_i^{re,lo}b_i^{re,hh}-\sum_{i=0}^{3}a_i^{im,lo}b_i^{im,hh}\Big)+2^{12}\Big(\sum_{i=0}^{3}a_i^{re,hi}b_i^{re,lh}-\sum_{i=0}^{3}a_i^{im,hi}b_i^{im,lh}\Big)\\
&+2^{8}\Big(\sum_{i=0}^{3}a_i^{re,lo}b_i^{re,hl}-\sum_{i=0}^{3}a_i^{im,lo}b_i^{im,hl}\Big)+2^{8}\Big(\sum_{i=0}^{3}a_i^{re,hi}b_i^{re,ll}-\sum_{i=0}^{3}a_i^{im,hi}b_i^{im,ll}\Big)\\
&+2^{4}\Big(\sum_{i=0}^{3}a_i^{re,lo}b_i^{re,lh}-\sum_{i=0}^{3}a_i^{im,lo}b_i^{im,lh}\Big)+\Big(\sum_{i=0}^{3}a_i^{re,lo}b_i^{re,ll}-\sum_{i=0}^{3}a_i^{im,lo}b_i^{im,ll}\Big)\\
M^{im}=\;&2^{20}\Big(\sum_{i=0}^{3}a_i^{re,hi}b_i^{im,hh}+\sum_{i=0}^{3}a_i^{im,hi}b_i^{re,hh}\Big)+2^{16}\Big(\sum_{i=0}^{3}a_i^{re,hi}b_i^{im,hl}+\sum_{i=0}^{3}a_i^{im,hi}b_i^{re,hl}\Big)\\
&+2^{12}\Big(\sum_{i=0}^{3}a_i^{re,lo}b_i^{im,hh}+\sum_{i=0}^{3}a_i^{im,lo}b_i^{re,hh}\Big)+2^{12}\Big(\sum_{i=0}^{3}a_i^{re,hi}b_i^{im,lh}+\sum_{i=0}^{3}a_i^{im,hi}b_i^{re,lh}\Big)\\
&+2^{8}\Big(\sum_{i=0}^{3}a_i^{re,lo}b_i^{im,hl}+\sum_{i=0}^{3}a_i^{im,lo}b_i^{re,hl}\Big)+2^{8}\Big(\sum_{i=0}^{3}a_i^{re,hi}b_i^{im,ll}+\sum_{i=0}^{3}a_i^{im,hi}b_i^{re,ll}\Big)\\
&+2^{4}\Big(\sum_{i=0}^{3}a_i^{re,lo}b_i^{im,lh}+\sum_{i=0}^{3}a_i^{im,lo}b_i^{re,lh}\Big)+\Big(\sum_{i=0}^{3}a_i^{re,lo}b_i^{im,ll}+\sum_{i=0}^{3}a_i^{im,lo}b_i^{re,ll}\Big),\\
E=\;&E_a+E_b
\end{aligned}}$$